print()

In [1]:
```
print("Darshan University","Rajkot", 360004)
```

```
Darshan University Rajkot 360004
```

In [3]:
```
print("Darshan","University","Rajkot",sep = '_')
```

```
Darshan_University_Rajkot
```

In [5]:
```
print(27,11,2025,sep = '/')
```

```
27/11/2025
```

In [7]:
```
print("Hello", end = '_')
print("Students", end = '*')
print("how are you?", end= '&')
```

```
Hello_Students*how are you?&
```

In [9]:
```
print("Welcome to Semester-4")
```

```
Welcome to Semester-4
```

In [11]:
```
fp = open("file_1.txt","w") #opens a text file named file_1 for writing
print("Welcome to Semester-4", file = fp)
fp.close()
```

In [13]:
```
name = "Priya"
spi = 9.5
```

In [15]:
```
print(name,"has got",spi,"SPI.")
```

```
Priya has got 9.5 SPI.
```

In [17]:
```
# her favourite subject is "Python".
```

In [19]:
```
print("her favourite subject is \"Python\".") #escape character
```

```
her favourite subject is "Python".
```

In [124…
```python
print('her favourite subject is "Python".')
```

```
her favourite subject is "Python".
```

In [21]:
```python
# line-1
# line-2
# line-3
```

In [23]:
```python
print("""line-1
line-2
line-3""") #multi line
```

```
line-1
line-2
line-3
```

f-string

In [26]:
```python
name = "Priya"
spi = 9.5
```

In [28]:
```python
print(f"{name} has got {spi} SPI.")
```

```
Priya has got 9.5 SPI.
```

In [30]:
```python
a = 5
b = 20
```

In [32]:
```python
print(f"Multiplication of {a} and {b} is : {a*b}")
```

```
Multiplication of 5 and 20 is : 100
```

In [34]:
```python
# Item      Price
# Pen       25
# Notebook  20
```

In [38]:
```python
print(f"{'Item':>10}:Price") # > for right alignment
print(f"{'Pen':^10}:25")     # ^ for center alignment
```

```
print(f"{'Notebook':10}:20") # default left alignment ( < )
```

```
      Item:Price
   Pen    :25
Notebook  :20
```

input()

In [41]:
```
# printf("Enter your name:");
# scanf("%s",name);
```

In [47]:
```
name = input("Enter your name:") #default type : String
print(name)
print(type(name))
```

```
Riya
<class 'str'>
```

In [49]:
```
name = float(input("Enter your SPI:")) #type-casting
print(name)
print(type(name))
```

```
8.9
<class 'float'>
```

.split()

In [54]:
```
s1 = "Priya,9.5"
```

In [56]:
```
n, s = s1.split(',')
s = float(s)
```

In [62]:
```
print(n)
```

```
Priya
```

In [64]:
```
print(type(s))
```

```
<class 'float'>
```

In [68]:
```
fn,ln,sem,cgpa = input("Enter your data(comma seperated):").split(",")
sem = int(sem)
cgpa = float(cgpa)
print(type(sem))
print(type(cgpa))
```

```
<class 'int'>
<class 'float'>
```

Datatype

int

In [72]:
```
i = 1_23_456
print(type(i))
```

```
<class 'int'>
```

In [74]:
```
print(i)
```

```
123456
```

In [76]:
```
import sys
```

In [78]:
```
sys.getsizeof(i)
```

Out[78]:
```
28
```

In [80]:
```
j = 123_
```

```
  Cell In[80], line 1
    j = 123_
           ^
SyntaxError: invalid decimal literal
```

In [82]:
```
k = _123
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[82], line 1
----> 1 k = _123

NameError: name '_123' is not defined
```

float

In [85]:
```
f = 1_2.2_5
```

In [87]:
```
print(f)
```

```
12.25
```

In [89]:
```
f = 1225e-2
```

In [91]:
```
print(f)
```

```
12.25
```

complex

In [94]:
```
c = 10 + 25j
```

In [96]:
```
print(type(c))
```

```
<class 'complex'>
```

In [98]:
```
print(c)
```

```
(10+25j)
```

In [100…
```
a = 45
b = 89
c1 = complex(a,b)
print(c1)
```

```
(45+89j)
```

In [102…
```
c1.real
```
Out[102…
```
45.0
```
In [104…
```
c1.imag
```
Out[104…
```
89.0
```

boolean

In [107…
```
b = False
```
In [109…
```
print(type(b))
```
```
<class 'bool'>
```
In [111…
```
b = True
print(type(b))
```
```
<class 'bool'>
```
In [113…
```
b = false
print(type(b))
```
```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[113], line 1
----> 1 b = false
      2 print(type(b))

NameError: name 'false' is not defined
```

string

In [116…
```
s1 = "Darshan"
print(type(s1))
```
```
<class 'str'>
```

In [118…

```python
s1 = 'Darshan'
print(type(s1))
```

```
<class 'str'>
```

In [120…

```python
s1 = '''Darshan University
Rajkot
Gujarat.'''
print(type(s1))
```

```
<class 'str'>
```

In [122…

```python
s1 = """Darshan University
Rajkot
Gujarat."""
print(type(s1))
```

```
<class 'str'>
```

list, tuple, set & dictionary

In [144…

```python
l1 = [10, 25.3, 6+5j, "Hi"]
print(l1)
print(type(l1))
```

```
[10, 25.3, (6+5j), 'Hi']
<class 'list'>
```

In [146…

```python
t1 = (10, 25.3, 6+5j, "Hi")
print(t1)
print(type(t1))
```

```
(10, 25.3, (6+5j), 'Hi')
<class 'tuple'>
```

In [148…

```python
s1 = {10, 25.3, 6+5j, "Hi"}
print(s1)
print(type(s1))
```

```
{25.3, 10, (6+5j), 'Hi'}
<class 'set'>
```

In [150…

```
d1 = {1:"One", 2:"Two", 3:"Three"}
print(d1)
print(type(d1))
```

```
{1: 'One', 2: 'Two', 3: 'Three'}
<class 'dict'>
```

Keywords in Python

In [129…

```
import keyword
print(keyword.kwlist)
print(type(keyword.kwlist))
print(len(keyword.kwlist))
```

```
['False', 'None', 'True', 'and', 'as', 'assert', 'async', 'await', 'break', 'class', 'continue', 'def', 'del', 'elif', 'else',
'except', 'finally', 'for', 'from', 'global', 'if', 'import', 'in', 'is', 'lambda', 'nonlocal', 'not', 'or', 'pass', 'raise',
'return', 'try', 'while', 'with', 'yield']
<class 'list'>
35
```

In [131…

```
int = 10
```

In [133…

```
int(45.23)
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[133], line 1
----> 1 int(45.23)

TypeError: 'int' object is not callable
```

In [135…

```
list = 45
```

In [137…

```
list((1,2,3,4))
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
Cell In[137], line 1
----> 1 list((1,2,3,4))

TypeError: 'int' object is not callable
```

In [139…
```
%reset
```

In [141…
```
int(45.23)
```

Out[141…
```
45
```

Python Operators

Arithmetic

In [158…
```
a = 10
b = 4
```

In [160…
```
print(a + b)
```

```
14
```

In [162…
```
print(a - b)
```

```
6
```

In [164…
```
print(a * b)
```

```
40
```

In [166…
```
print(a / b)
```

```
2.5
```

In [168…
```
print(a // b)
```

```
2
```

In [170…
```python
print(a ** b)
```

```
10000
```

In [172…
```python
print(a % b)
```

```
2
```

Assignment

In [175…
```python
# a = a + 1
a += 1
```

In [177…
```python
print(a)
```

```
11
```

Logical (and, or , not)

In [180…
```python
True and True
```

Out[180…
```
True
```

In [182…
```python
True and False
```

Out[182…
```
False
```

In [184…
```python
False and False
```

Out[184…
```
False
```

In [187…
```python
True or False
```

Out[187…
```
True
```

In [189…
```python
False or False
```

Out[189… False

In [191…
```
True or False
```

Out[191… True

In [193…
```
not False
```

Out[193… True

In [195…
```
not True
```

Out[195… False

Relational

In [210…
```
print(a, b)
```

```
11 4
```

In [198…
```
a > b
```

Out[198… True

In [200…
```
a < b
```

Out[200… False

In [202…
```
a >= b
```

Out[202… True

In [204…
```
a <= b
```

Out[204… False

In [206…
```
a == b
```

Out[206…

```
False
```

In [208…

```
a != b
```

Out[208…

```
True
```

bitwise

In [215…

```
a = 10
b = 4
```

In [217…

```
print(a, b)
```

```
10 4
```

In [221…

```
bin(a)
```

Out[221…

```
'0b1010'
```

In [223…

```
bin(b)
```

Out[223…

```
'0b100'
```

In [249…

```
a & b #bitwise and
```

Out[249…

```
0
```

In [251…

```
a | b #bitwise or
```

Out[251…

```
14
```

In [253…

```
a ^ b #bitwise xor
```

Out[253…

```
14
```

In [255…

```
~a #bitwise nor
```

Out[255… -11

In [259…
```
a >> 1 #left shift by one -> divides the number by 2^1
```

Out[259… 5

In [263…
```
a >> 2 #left shift by two -> divides the number by 2^2
```

Out[263… 2

In [265…
```
a >> 3 #left shift by three -> divides the number by 2^3
```

Out[265… 1

In [269…
```
a >> 4 #left shift by four -> divides the number by 2^4
```

Out[269… 0

In [271…
```
a >> 5 #left shift by five -> divides the number by 2^5
```

Out[271… 0

In [273…
```
a << 1 #right shift by one -> multiplies the number by 2^1
```

Out[273… 20

In [275…
```
a << 2 #right shift by two -> multiplies the number by 2^2
```

Out[275… 40

In [277…
```
a << 3 #right shift by three -> multiplies the number by 2^3
```

Out[277… 80

Conditional Statements

simple if

In [281…

```
num = 15
```

In [283…

```
if num > 10:
    print(f"{num} is greater than 10.")
```

```
15 is greater than 10.
```

if..else

In [286…

```
age = 16
if age >= 18:
    print("can vote")
else:
    print("can not vote.")
```

```
can not vote.
```

nested if

In [289…

```
age = 25
has_id = False
```

In [291…

```
if age >= 18:
    if has_id:
        print('Go and vote.')
    else:
        print("ID is required.")
else:
    print("Not eligible to vote.")
```

```
ID is required.
```

elif ladder

In [294…

```
age = 3
has_id = False
if age >= 18 and has_id:
    print('Go and vote.')
elif age >= 18 and not has_id:
    print("ID is required.")
else:
    print("Not eligible to vote.")
```

```
Not eligible to vote.
```

In [296…

```
name = input("Enter your name:")
marks = int(input("Enter your marks :"))
if marks >= 80:
    grade = "A"
elif marks >= 70:
    grade = "B"
elif marks >= 60:
    grade = "C"
elif marks >= 50:
    grade = "D"
else:
    grade = "F"
print(f"{name} has got - {grade} Grade.")
```

```
Riya has got - F Grade.
```

match - case

In [300…

```
no_of_day = 1
match no_of_day:
    case 1:
        print("Monday")
    case 2:
        print("Tueday")
    case 3:
```

```
        print("Wednesday")
    case 4:
        print("Thursday")
    case 5:
        print("Friday")
    case 6:
        print("Saturday")
    case 7:
        print("Sunday")
    case _:
        print("Invalid Input")
```

```
Monday
```

In [302…

```
no_of_day = 1
match no_of_day:
    case 1 | 2 | 3 | 4 | 5:
        print("Week days")
    case 6 | 7:
        print("Weekend")
    case _:
        print("Invalid Input")
```

```
Week days
```

Ternary Operator

In [ ]:

```
# a = 15
# b = 20
# if a > b:
#     max = a
# else:
#     max = b
```

In [308…

```
# () ? T : F # in c - java
```

In [310…

```
# True statement if condition else False statement
```

In [312…

```
a = 15
```

```
b = 20
```

In [314…

```
res = a if a > b else b
print(res)
```

```
20
```

In [316…

```
print(a) if a > b else print(b)
```

```
20
```

In [318…

```
print(a if a > b else b)
```

```
20
```

nesting of ternary operator

In [321…

```
a, b, c = 11, 12, 13
```

In [323…

```
if a > b and a > c:
    res = a
elif b > a and b > c:
    res = b
else:
    res = c
```

In [325…

```
if a > b and a > c:
    res = a
else:
    if b > c:
        res = b
    else:
        res = c
```

In [327…

```
res = a if a > b and a > c else b if b > c else c
```

In [329…

```
res
```

Out[329…

```
13
```

In [331…
```
res = a if a > c else c if a > b else b if b > c else c
```

In [333…
```
res
```

Out[333… 13

In [337…
```
ans = (a if a > b else b) if (a if a > b else b) > c else c
```

In [339…
```
ans
```

Out[339… 13

looping statements

for loop

range()

range() is a built-in function used to generate a sequence of numbers.

It is commonly used in loops, especially for loops.

range() does not create a list, It creates a range object.

range(stop)

Starts from 0 and goes up to stop-1.

In [349…
```
for i in range(5):
    print(i, end = "\t")
```

```
0	1	2	3	4
```

range(start, stop)

starts from start and goes up to stop-1. default start = 0

In [354…

```
for i in range(2, 7):
    print(i, end = '\t')
```

```
2	3	4	5	6
```

range(start, stop, step)

Step defines how much to jump each time. default step = 1

In [357…

```
for i in range(1, 10, 2):
    print(i, end = "\t")
```

```
1	3	5	7	9
```

Negative step (reverse counting)

In [360…

```
for i in range(10, 0, -1):
    print(i, end = "\t")
```

```
10	9	8	7	6	5	4	3	2	1
```

factors of a given number

In [373…

```
num = 25
for i in range(1, num+1):
    if num % i == 0:
        print(i)
```

```
1
5
25
```

break, continue & pass

In [376…

```
for i in range(5):
    if i == 3:
        break
    print(i)
```

```
0
1
2
```

In [378…

```
for i in range(5):
    if i == 3:
        continue
    print(i)
```

```
0
1
2
4
```

In [380…

```
for i in range(5):
    pass
```

for...else

In [383…

```
num = 11
for i in range(2,num):
    if num % i == 0:
        print("Not Prime")
        break
else:
    print("Prime")   #else block executes only when the for loop executes completely
                     #without executing break statement
```

```
Prime
```

nested for loop

prime numbers between 1 to 100

In [387…

```
for n in range(1, 101):
    for i in range(2,n):
        if n % i == 0:
            break
    else:
        print(n, end = "\t")
```

```
1       2       3       5       7       11      13      17      19      23      29      31      37      41      43      47
53      59      61      67      71      73      79      83      89      97
```

matrix like printing

In [390…

```
for i in range(3):
    for j in range(4):
        print(f"({i}-{j})", end = "\t")
    print()
```

```
(0-0)   (0-1)   (0-2)   (0-3)
(1-0)   (1-1)   (1-2)   (1-3)
(2-0)   (2-1)   (2-2)   (2-3)
```

while loop

In [397…

```
i = 1
while i <= 10:
    print(i, end = "\t")
    i += 1   # i++ is not supported
```

```
1       2       3       4       5       6       7       8       9       10
```

In [401…

```
i = 1
while i <= 10:
    print(i, end = "\t")
    i *= 2
```

```
1       2       4       8
```

In [403…

```python
num = 1234
while num > 0:
    r = num % 10
    print(r)
    num //= 10   # floor division
```

```
4
3
2
1
```

In [405…

```python
num = 1234
sum_digit = 0
while num > 0:
    r = num % 10
    sum_digit += r
    num //= 10   # floor division
print(sum_digit)
```

```
10
```

In [411…

```python
num = 1234
n = num
rev = 0
while num > 0:
    r = num % 10
    rev = (10 * rev) + r
    num //= 10   # floor division
print(rev)
```

```
4321
```

In [413…

```python
if n == rev:
    print(f"{n} is palindrome number.")
else:
    print(f"{n} is not palindrome number.")
```

```
1234 is not palindrome number.
```